International Trade

Problem Set

Multiple-Choice Questions

(1). Which of the following statements is **NOT** true?

A. The slope of the PPF in the Ricardian model is the ratio of the marginal products of labor in producing two goods;
B. Comparative advantage can also be interpreted as lower opportunity cost;
C. Countries can specialize according to their comparative advantage and achieve a higher level of utility after opening to trade.
D. Every country must have its own comparative advantage and absolute advantage.

(2). In the **short run**, an outflow of capital is expected to cause a(n) _____in the real rental of capital and a(n) _____in the real rental of land in the origin country.

A. decrease; decrease
B. increase; decrease
C. decrease; increase
D. increase; increase

(3). Consider a natural disaster that destroyed half of the agricultural land in the US. Using the specific-factors model with two sectors, manufacturing and agricultural, which of the following is **NOT** likely to happen?

A. The wage of workers will decrease.
B. Labor will be moving away from agricultural sector into the manufacturing sector.
C. The rental of capital will increase.
D. The rental of land will decrease.

(4). In the short-run specific-factor model, assume new farming land is discovered. Which of the following is **NOT** likely to happen (assuming nominal prices stay the same):

A. The real rental of capital will decrease.
B. The real wage will increase, labor is better off.
C. The real rental of land will increase.
D. Agricultural output will increase.

(5). Which of the following statements about trade policy is **NOT** true?

A. The optimal import tariff is always positive for not-small economies.
B. No matter it is perfect competition or imperfect competition in the foreign exporting country, the importing country (if large enough) can always benefit from adopting some level of tariff.
C. Import tariff and import quota are equivalent under perfect market competition.
D. It is never welfare-improving to adopt voluntary export restraint.

Question 1. True or False. Briefly explain your judgements.

a. If there are two goods, Apples (A) and Bananas (B), then Home has a comparative advantage in the production of apples if $P_A < P_A^W$ and $P_B > P_B^W$ (where P_A and P_B are domestic prices and P_A^W and P_B^W are world prices).

b. Land and capital owners usually welcome foreign immigrants.

c. Factors that can be used/employed in the exporting industries will always gain from trade liberalization.

d. In the Heckscher-Ohlin model, when a labor abundant country opens to trade with the world, its wage rental ratio goes up and every producer demands a higher labor-capital ratio.

e. In the long run, international movement of labor and capital will not affect the real wage and real capital rental across countries, but it will affect the output of goods produced across countries.

f. In the short run, an outflow of capital is expected to cause an increase in the real rental of capital and an increase in the real rental of land in the origin country.

g. As China receives more and more FDI, under the assumptions of Heckscher-Ohlin model, it will produce more and more capital-intensive products.

h. As the labor cost increases in China, more activities will be offshored to other developing countries such as Vietnam who has comparative advantage in labor-intensive tasks. This type of offshoring will benefit the workers in Vietnam and the capital owners in China will also benefit.

Question 2. In Singapore, it takes 4 months for 15 workers to make 100 tons of oil. It also requires 60 workers to work 1 year to produce 6 million micro conductors; In Malaysia, 10 workers can make 80 tons of oil in 6 months. And it requires 100 workers working 1 year to produce 4 million micro conductors. Assume constant marginal production of labor. Determine the comparative advantage in oil and micro conductors for Malaysia and Singapore.

Question 3. There are two sectors, all Industries (I) and Apparel (A). Think of Bangladesh as Home and China as Foreign. Examine the table below and answer the questions:

	Bangladeshi Output per Worker as % of China	Bangladeshi exports as % of China
All industries	28.5	1.0
Apparel	77	15.5

Source: McKinsey and Company, "Bangladesh's ready-made garments industry: The challenge of growth," 2012; UN Monthly Bulletin of Statistics.

a. Determine the comparative advantage in the two sectors for the two countries.
b. Is the data in line with or against the Ricardian model?

Question 4. Colombia is one of the richest countries in terms of natural resources. It exports products like coffee and is the number two supplier of flowers in the world. Consider the specific factors model with two goods: coffee (C) and flowers (F), and assume Colombia has comparative advantage in producing coffee. Coffee is produced using land T and capital K, while flowers are made with land T and labor L. Answer the following questions:

a. In this version of the model, what are the specific factors? Which factor is the mobile factor?
b. How is the real rental of land affected by opening to trade?
c. How do the nominal and real wages of workers change?
d. Suppose that after Columbia has opened to trade, it receives foreign direct investment (i.e., an increase in K). Discuss the short-run effects on the quantities of two goods produced, the real rental of land, the real rental of capital and the real wage.

Question 5. In the Heckscher-Ohlin model, assume home is capital abundant. There are two goods: computer and shoe. Computer is capital-intensive and shoe is labor-intensive. Starting from the free trade equilibrium, if we downsize the home country's endowment level (shrink the endowment of capital and labor proportionally).

a. Discuss its effect on the relative price of computer.
b. Will labor or capital be better off due to this endowment change?
c. What is the effect on labor/capital ratio in computer production?
d. Will it change the trade pattern (or comparative advantage) between Home and Foreign?